## IV. Match the following:

| | Column - I | | Column - II |
|---|---|---|---|
| 1 | Retina | a | Path way of light |
| 2 | Pupil | b | Far point comes closer |
| 3 | Ciliary muscles | c | near point moves away |
| 4 | Myopia | d | Screen of the eye |
| 5 | Hypermetropia | f | Power of accom-modation |

## V. Assertion and reasoning type

Mark the correct choice as

a) If both assertion and reason are true and reason is the correct explanation of assertion.

b) If both assertion and reason are true but reason is not the correct explanation of assertion.

c) Assertion is true but reason is false.

d) Assertion is false but reason is true.

1. **Assertion:** If the refractive index of the medium is high (denser medium) the velocity of the light in that medium will be small

   **Reason:** Refractive index of the medium is inversely proportional to the velocity of the light

2. **Assertion:** Myopia is due to the increase in the converging power of eye lens.

   **Reason:** Myopia can be corrected with the help of concave lens.

## VI. Answer Briefly

1. What is refractive index?
2. State Snell's law.
3. Draw a ray diagram to show the image formed by a convex lens when the object is placed between F and 2F.
4. Define dispersion of light
5. State Rayleigh's law of scattering
6. Differentiate convex lens and concave lens.
7. What is power of accommodation of eye?
8. What are the causes of 'Myopia'?
9. Why does the sky appear in blue colour?
10. Why are traffic signals red in colour?

## VII. Give the answer in detail

1. List any five properties of light
2. Explain the rules for obtaining images formed by a convex lens with the help of ray diagram.
3. Differentiate the eye defects: Myopia and Hypermetropia
4. Explain the construction and working of a 'Compound Microscope'.

## VIII. Numerical Problems:

1. An object is placed at a distance 20cm from a convex lens of focal length 10cm. Find the image distance and nature of the image.
2. An object of height 3cm is placed at 10cm from a concave lens of focal length 15cm. Find the size of the image.

## IX. Higher order thinking (HOT) questions:

1. While doing an experiment for the determination of focal length of a convex lens, Raja Suddenly dropped the lens. It got broken into two halves along the axis. If he continues his experiment with the same lens, (a) can he get the image? (b) Is there any change in the focal length?
2. The eyes of the nocturnal birds like owl are having a large cornea and a large pupil. How does it help them?

## REFERENCE BOOKS


1. Fundamentals of optics by D.R. Khanna and H.R. Gulati, R. Chand & Co.
2. Principles of Physics – Halliday, Resnick & Walker, Wiley Publications, New Delhi.


## INTERNET RESOURCES

1. www.physicsabout.com
2. www.khanacademy.org



10th_Science_Unit-2.indd 30 12/2/2022 4:59:18 PM